

E-Commerce Microservices Backend System

Java 17 | Spring Boot | Spring Cloud | Kafka | JWT | Eureka | API Gateway | MySQL

Services

- User Service
- Product Service
- Order Service
- Notification Service

Platform

- API Gateway
- Eureka Server
- Feign Client

Security & Events

- JWT Authentication
- Spring Security
- Apache Kafka

Project Overview

A backend project demonstrating distributed microservices design and API-driven development.

Architecture

- Microservices-based backend
- Centralized routing via API Gateway
- Service discovery using Eureka

Business Flow

- User registration and login
- Product and inventory management
- Order creation and notification flow

Integration

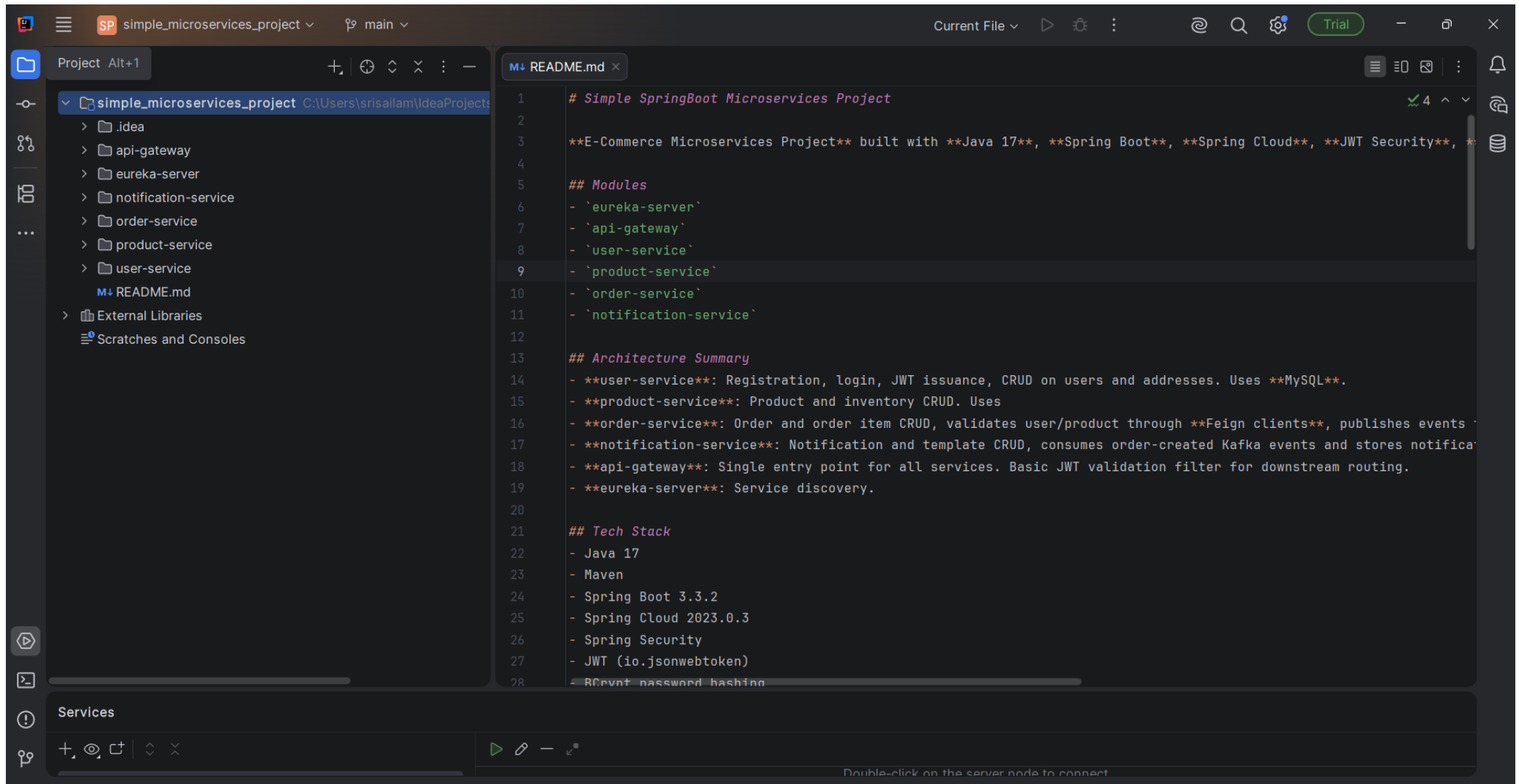
- Feign Client for service calls
- Kafka event on order creation
- Separate MySQL databases per service

System Flow

- Client requests enter through API Gateway
- Gateway routes requests to User/Product/Order/Notification services
- Order Service publishes order-created events to Kafka
- Notification Service consumes events and stores notification data in MySQL

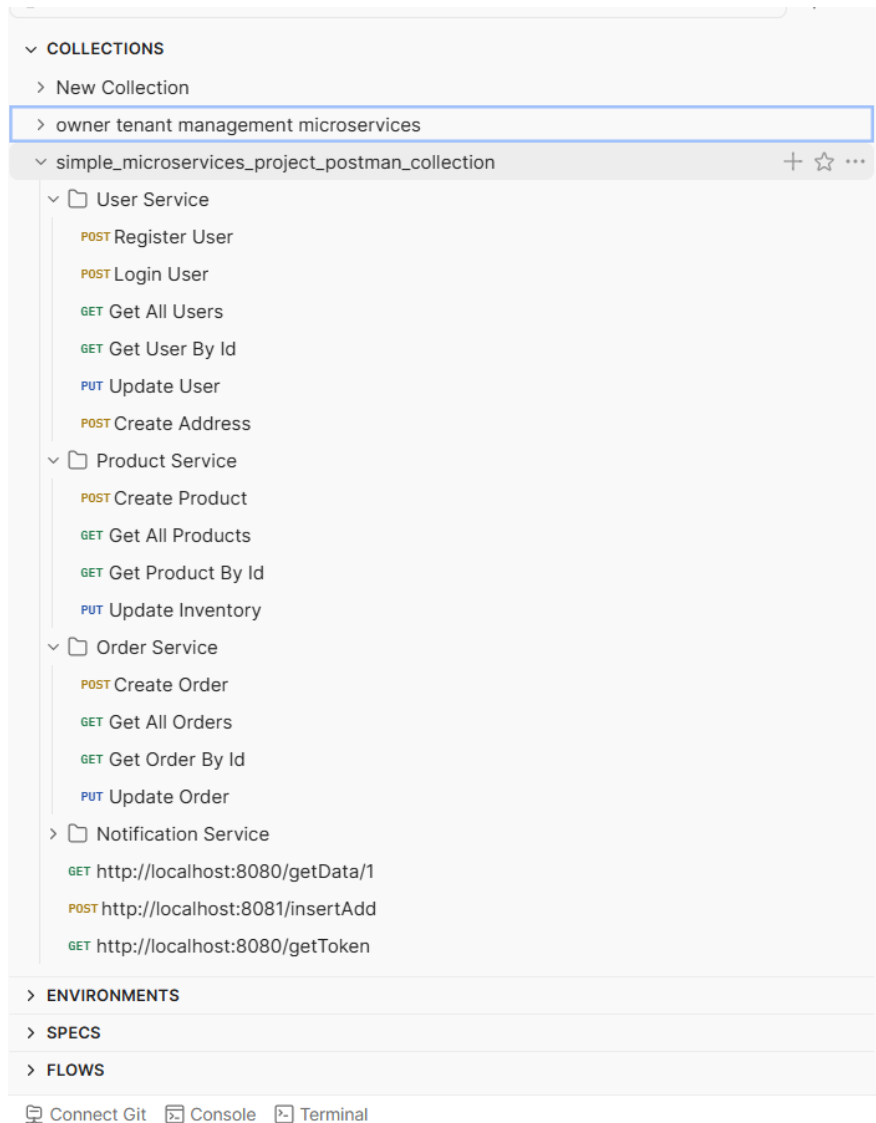
IntelliJ IDEA Project Structure

Multi-module Spring Boot project containing API Gateway, Eureka Server, and service modules.



Postman API Collection - Service-wise REST Endpoints

Organized API collection covering user, product, order, and notification service endpoints.



User Service APIs

- Register User
- Login User
- Get/Update User
- Create Address

Product Service APIs

- Create Product
- Get Products
- Get Product By Id
- Update Inventory

Order Service APIs

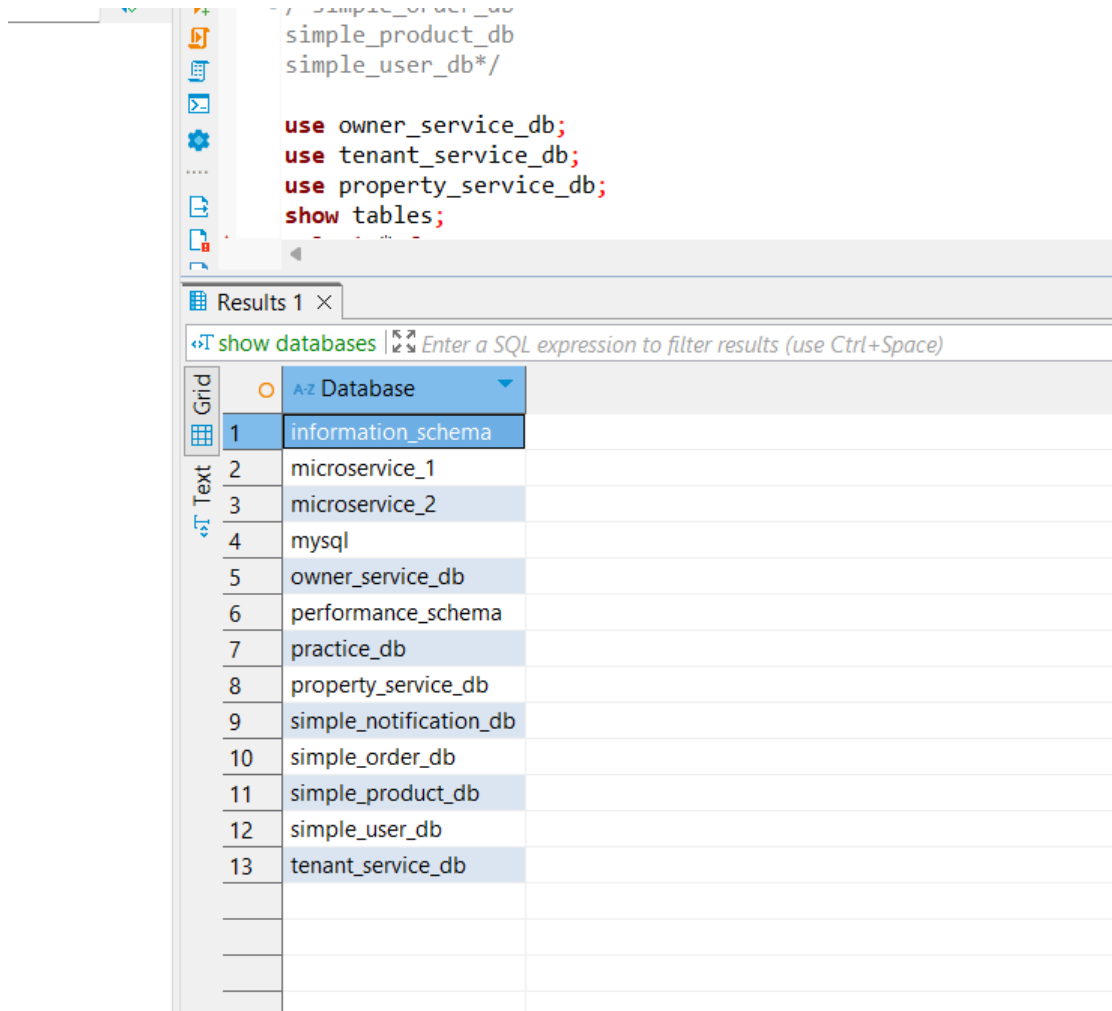
- Create Order
- Get All Orders
- Get Order By Id
- Update Order

Notification Service

- Notification endpoints
- Token and data APIs
- Kafka consumer flow support

MySQL Service Databases

Separate local databases created for microservice data ownership and persistence.



Databases Used

- simple_user_db
- simple_product_db
- simple_order_db
- simple_notification_db

Design Note

- Each microservice owns its database
- Data access through JPA/Hibernate
- Supports independent service development

README & Tech Stack Summary

Clean project documentation for recruiters and interviewers reviewing the GitHub repository.

Core Modules

- eureka-server
- api-gateway
- user-service
- product-service
- order-service
- notification-service

Tech Stack

- Java 17
- Spring Boot 3.3.2
- Spring Cloud 2023.0.3
- Spring Security + JWT
- Spring Data JPA + MySQL
- Apache Kafka
- Feign Client
- Maven

Run Flow

- Start MySQL
- Start Kafka KRaft
- Start Eureka Server
- Start API Gateway
- Start all services
- Test APIs in Postman